

E088

Natural Language Processing for AI Engineers

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A support-ticket classifier at a gadget company with a flagship store in Seattle uses a word-level vocabulary built two years ago. New product names now show up as [UNK] and accuracy on those tickets drops. The company releases products every spring. What is the best fix?

- A. Lowercase all text.
- B. Ignore [UNK] tokens; unknown words never matter.
- C. Add product names to the stopword list.
- D. Switch to a subword tokenizer and retrain, using the same tokenizer at inference.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A review-sentiment pipeline at a restaurant-booking site with a green logo removes standard English stopwords, including “not.” The site lists 20,000 restaurants. Reviews like “not good at all” are being scored positive. What should change?

- A. Keep negation words; do not remove them for sentiment.
- B. Stopword removal always improves accuracy, so keep it.
- C. Remove even more stopwords.
- D. Lemmatize the remaining words.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

For which task is removing common stopwords most reasonable?

- A. Answering questions such as “Which drugs should not be combined?”
- B. Grouping news articles by topic with a bag-of-words model.
- C. Sentiment analysis of reviews.
- D. Machine translation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A search team at a travel site with a popular newsletter compares two query embeddings, $a = (1, 2, 2)$ and $b = (2, 1, 2)$. The embeddings were trained on hotel reviews. What is their cosine similarity?

- A. About 0.89
- B. 8
- C. About 0.11
- D. About 2.67

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which embedding approach gives “bank” different vectors in “river bank” and “bank loan”?

- A. One-hot encoding
- B. Contextual embeddings from a transformer encoder
- C. Static word2vec embeddings
- D. Any embedding, because all embeddings capture context.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A legal-tech firm with a large conference booth feeds 3,000-token contracts into a transformer classifier with a 512-token limit. Key clauses often appear near the end. The firm uses a cloud GPU. What should the team do?

- A. Increase the batch size.
- B. Switch to a bag-of-words model.
- C. Split contracts into overlapping chunks and combine the predictions, or use a longer-context model.
- D. Nothing; attention sees the whole document regardless of length.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A help desk at a university with 30,000 students receives 1,000 tickets; 50 are urgent. The classifier, built during a hackathon, flags 40 as urgent: 30 correctly and 10 incorrectly, missing 20 urgent tickets. What is the F1 score for the urgent class?

- A. 0.97
- B. 0.45
- C. 0.675
- D. About 0.67

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A classifier labels every ticket “not urgent.” 5% of tickets are urgent. What are its accuracy and its F1 on the urgent class?

- A. 95% accuracy and F1 of 0.
- B. 95% accuracy and F1 of 0.95.
- C. 5% accuracy and F1 of 0.
- D. 95% accuracy and F1 of 0.5.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A sentiment model at a streaming service with a new comedy lineup scores 94% on its test set. Reviewers find it rates “I don’t love this show” as positive and fails on common typos. The test set was labeled in March. What is the best next step?

- A. Relabel the March test set.
- B. Retrain on a larger random sample with no error review.
- C. Build stress tests for negation, typos, and paraphrase, and group errors by type.
- D. Report the 94% score; that proves it is reliable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A toxicity filter at a social app with a purple theme reaches 91% overall accuracy. An audit shows it flags harmless posts written in one regional dialect far more often than others. The app has 2 million users. What should the team do?

- A. Remove posts in that dialect from the training data.
- B. Measure false-positive rates by dialect, improve data and thresholds, and document the gap.
- C. Bias only matters for generative models.
- D. Keep it; 91% overall accuracy shows it is fair.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Before launching a question-answering model, a health-insurance company with a new mobile app must produce a governance artifact. Its engineers use a shared code repository. Which content belongs in that model card?

- A. The full source code from the repository.
- B. The training hyperparameter log only.
- C. Intended and out-of-scope uses, evaluation results by slice, known limitations, and a monitoring plan.
- D. A marketing summary of the model's benefits.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A model card states: "Not for giving medical diagnoses." Which section does this belong in?

- A. Intended uses
- B. Training data
- C. Evaluation results
- D. Out-of-scope uses

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A support-ticket classifier at a gadget company ~~with a flagship store in Seattle~~ uses a word-level vocabulary built two years ago. New product names now show up as [UNK] and accuracy on those tickets drops. ~~The company releases products every spring.~~ What is the best fix?

Target word: handle unseen words **Decides:** Word-level vocabulary cannot represent new words.

A	Adjacent category	Does not add new words to the vocabulary.
B	Absolute language	Product names are often the key clue in tickets.
C	Inverted operation	Removes the very words that carry the signal.
D	Correct	New words split into known pieces instead of [UNK].

2. Answer A Concept 2

A review-sentiment pipeline at a restaurant-booking site ~~with a green logo~~ removes standard English stopwords, including “not.” ~~The site lists 20,000 restaurants.~~ Reviews like “not good at all” are being scored positive. What should change?

Target word: preprocessing fix **Decides:** Stopword removal deletes negations.

A	Correct	“Not” flips the meaning, so it must stay.
B	Absolute language	Its value depends on the task.
C	Inverted operation	Deletes more meaning-bearing words.
D	Adjacent category	Normalizes word forms but does not restore “not.”

3. Answer B Concept 2 · transfer

For which task is removing common stopwords most reasonable?

Target word: reverse: task where the step fits **Decides:** Function words carry little topic signal.

A	Confusion set	Function words change the question’s meaning.
B	Correct	Topic is carried by content words.
C	Inverted operation	Negations like “not” flip sentiment.
D	Adjacent category	Every word must be translated.

4. Answer A Concept 3

A search team at a travel site ~~with a popular newsletter~~ compares two query embeddings, $a = (1, 2, 2)$ and $b = (2, 1, 2)$. ~~The embeddings were trained on hotel reviews.~~ What is their cosine similarity?

Target word: cosine similarity **Decides:** $a \cdot b = 8$; $|a| = |b| = 3$.

A	Correct	$8 / (3 \times 3) = 8/9$.
B	Adjacent category	The raw dot product, not normalized.
C	Inverted operation	That is cosine distance, $1 - \text{similarity}$.
D	Dropped element	Divides by only one vector’s length.

5. Answer B Concept 3 · transfer

Which embedding approach gives “bank” different vectors in “river bank” and “bank loan”?

Target word: static vs. contextual **Decides:** Only contextual models vary vectors by sentence.

A	Confusion set	Same vector for every use and no meaning captured.
B	Correct	Each token’s vector depends on its surrounding words.
C	Inverted operation	One fixed vector per word type.
D	Absolute language	Static embeddings do not.

6. Answer C Concept 4

A legal-tech firm ~~with a large conference booth~~ feeds 3,000-token contracts into a transformer classifier with a 512-token limit. Key clauses often appear near the end. ~~The firm uses a cloud GPU.~~ What should the team do?

Target word: handle long inputs **Decides:** Input exceeds the context window; the end is truncated.

A	Confusion set	Batch size does not change sequence length.
B	Adjacent category	Reads all words but loses order and clause meaning.
C	Correct	Ensures the end clauses are actually read.
D	Absolute language	Tokens past 512 are cut off.

7. Answer D Concept 5

A help desk at a university ~~with 30,000 students~~ receives 1,000 tickets; 50 are urgent. The classifier, ~~built during a hackathon~~, flags 40 as urgent: 30 correctly and 10 incorrectly, missing 20 urgent tickets. What is the F1 score for the urgent class?

Target word: F1 for the minority class **Decides:** Precision = 30/40 = 0.75; recall = 30/50 = 0.60.

A	Confusion set	That is overall accuracy (970 of 1,000 correct).
B	Dropped element	$P \times R$ without the $2 / (P + R)$ factor.
C	Adjacent category	The simple average of precision and recall, not the harmonic mean.
D	Correct	$2(0.75)(0.60) / 1.35 \approx 0.667$.

8. Answer A Concept 5 · transfer

A classifier labels every ticket “not urgent.” 5% of tickets are urgent. What are its accuracy and its F1 on the urgent class?

Target word: accuracy vs. F1 on a trivial model **Decides:** It never flags an urgent ticket: recall = 0.

A	Correct	Right on every non-urgent ticket, useless on urgent ones.
B	Confusion set	Treats F1 as another name for accuracy.
C	Inverted operation	It is correct on the 95% majority.
D	Adjacent category	0.5 is chance-level thinking; with zero true positives F1 is 0.

9. Answer C Concept 6

A sentiment model at a streaming service ~~with a new comedy lineup~~ scores 94% on its test set. Reviewers find it rates “I don’t love this show” as positive and fails on common typos. ~~The test set was labeled in March.~~ What is the best next step?

Target word: find failure modes **Decides:** Failures appear under small, realistic input changes.

A	Camouflage	The label date is not the cause of these failures.
B	Dropped element	Skips learning what is failing and why.
C	Correct	Targeted tests reveal and track specific weaknesses.
D	Absolute language	One score hides specific failure modes.

10. Answer B Concept 7

A toxicity filter at a social app ~~with a purple theme~~ reaches 91% overall accuracy. An audit shows it flags harmless posts written in one regional dialect far more often than others. ~~The app has 2 million users.~~ What should the team do?

Target word: bias response **Decides:** Higher false-positive rate for one group.

A	Inverted operation	Less representation usually makes errors on that group worse.
B	Correct	Slice metrics expose the harm; fixes and documentation address it.
C	Confusion set	Classifiers can cause unequal harm too.
D	Dropped element	An overall number can hide a group’s high error rate.

11. Answer C Concept 8

Before launching a question-answering model, a health-insurance company ~~with a new mobile app~~ must produce a governance artifact. ~~Its engineers use a shared code repository.~~ Which content belongs in that model card?

Target word: model card contents **Decides:** Model cards record use, limits, evaluation, and monitoring.

A	Camouflage	Code is not a record of use, limits, or evaluation.
B	Dropped element	Leaves out intended use, limits, and monitoring.
C	Correct	These are the core sections of a model card.
D	Adjacent category	Promotional, not a governance record.

12. Answer D Concept 8 · transfer

A model card states: “Not for giving medical diagnoses.” Which section does this belong in?

Target word: reverse: statement to section **Decides:** It names a use the model must not serve.

A	Inverted operation	That section lists what the model should be used for.
B	Adjacent category	Describes data sources, not allowed uses.
C	Confusion set	Reports measured performance, not usage limits.
D	Correct	Lists uses the model is not designed or approved for.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Subword tokenization	Subword tokenizers (BPE, WordPiece) split rare or new words into known pieces, avoiding unknown tokens; the same tokenizer must be used in training and inference.	Accepting [UNK] for new words, or swapping tokenizers between training and serving.	New product names, misspellings, or rare words becoming unknown.
2	Preprocessing must fit the task	Steps like stopword removal or lowercasing can erase signal; negations matter for sentiment, while topic tasks can often drop function words.	Applying a standard cleaning recipe to every task.	Words such as “not” or “no” disappearing in preprocessing.
3	Embeddings and similarity	Cosine similarity = $a \cdot b / (a b)$; static embeddings give one vector per word, contextual embeddings change with the sentence.	Using the raw dot product as similarity, or expecting static vectors to separate word senses.	Two vectors to compare, or one word used in two senses.
4	Transformer context windows	Self-attention links any two tokens inside the window, but text beyond the maximum length is truncated unless chunked or handled by a longer-context model.	Assuming a transformer reads a document of any length.	Documents longer than a model’s token limit.
5	Metrics beyond accuracy	With imbalanced classes, use precision, recall, and $F1 = 2PR / (P + R)$ for the class that matters; accuracy can look high while the key class is missed.	Reporting overall accuracy on a rare-class problem.	A small minority class and a high accuracy number.
6	Error analysis and stress testing	Group errors by type and test behavior under realistic changes (negation, typos, paraphrase) to find failure modes a single score hides.	Treating one held-out score as proof of reliability.	A model that fails on small wording changes.
7	Bias assessment across groups	Measure error rates, especially false positives and negatives, for each dialect or demographic slice and act on gaps.	Declaring a model fair because overall accuracy is high.	One group’s text being flagged or missed more often.
8	Governance artifacts (model cards)	A model card documents intended use, out-of-scope uses, evaluation results by slice, limitations, and monitoring expectations.	Treating code, logs, or marketing copy as the governance record.	A request to document how a model should and should not be used.

Course scope (public description): This course develops students’ ability to apply natural language processing (NLP) techniques to solve engineering problems involving human language data. Learners implement reproducible text preprocessing and tokenization pipelines, apply word embeddings, and use sequence-based and transformer-based models to complete tasks such as classification, sentiment analysis, and question answering. The course emphasizes rigorous evaluation beyond accuracy, using task-appropriate metrics, structured error analysis, and stress testing under realistic input variation, to identify failure modes and reliability risks. Learners

also assess bias, safety, and responsible-use concerns (including generative AI risks when applicable) and produce governance artifacts that document intended use, limitations, evaluation results, and monitoring expectations. Through applied engineering work, learners demonstrate judgment in selecting methods and designing solutions that meet technical, ethical, and operational constraints.